

Report R14 — one figure for the WCC/SCC planes and the flow between them

Five candidate layouts

Tier 1 analysis

2026-08-04

R9 spends three figures on a single idea: every rule has a position in the sector plane, a position in the monitored-attractor plane, and dissipation moves it from the first to the second. Five condensed layouts follow, all from the same data (`obc0`, 254 rules with both descriptors) and the same design rules — marker area grows with the number of rules stacked in a cell, colour is the family, arrows are light and thicken with the number of rules making the same move, and a rule that does not move gets no arrow. Only the layout differs.

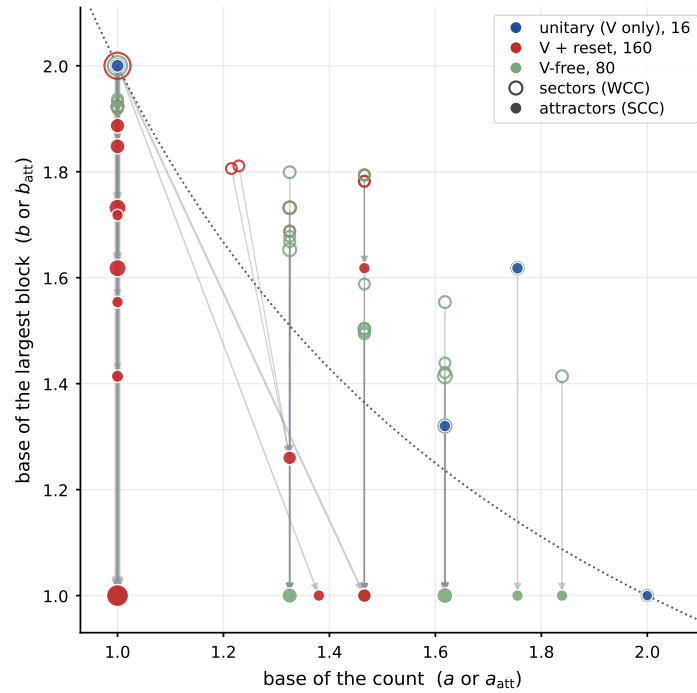


Figure 1: **A — overlay.** One plane; each rule appears twice, hollow where its sectors put it and filled where its attractors do, joined by an arrow. The only layout in which both coordinates live on *the same axes*, so the move reads as a displacement rather than a mapping between two spaces — and the one that makes the dominant fact visible: of the 230 rules that move, 96% move **straight down** — a unchanged, b collapsed — and *all* 230 decrease b . Not one moves horizontally. Dissipation destroys block size and leaves the count alone. Most compact of the five.

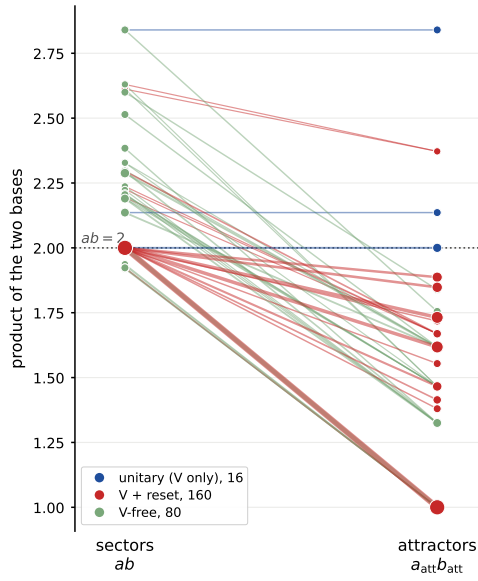


Figure 2: **B — slopegraph.** Neither coordinate, but the product ab that the exclusion curve constrains. Sectors left, attractors right, one line per rule. Discards the a/b split and keeps only what the theorem is about, which makes the theorem the picture: everything starts on or above 2, the unitary rules run exactly horizontal, and almost everything else drops below.

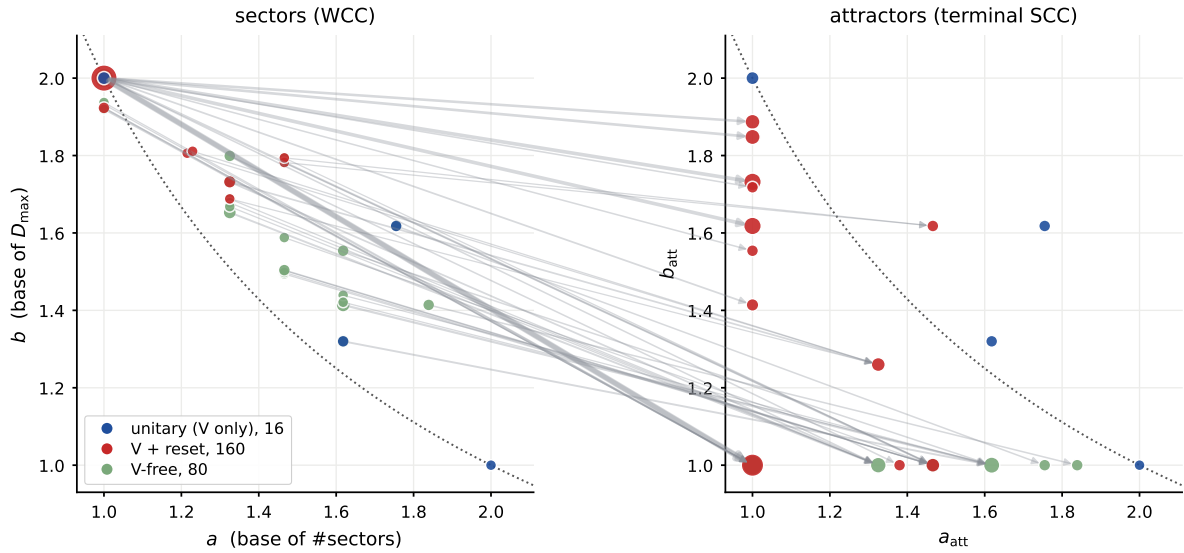


Figure 3: **C — twin panels, horizontal.** The literal reading: sectors left, attractors right, arrows across the gap. Easiest to caption and the least ambiguous about which plane is which, but the arrows bundle in the gutter and the gap costs about a third of the width.

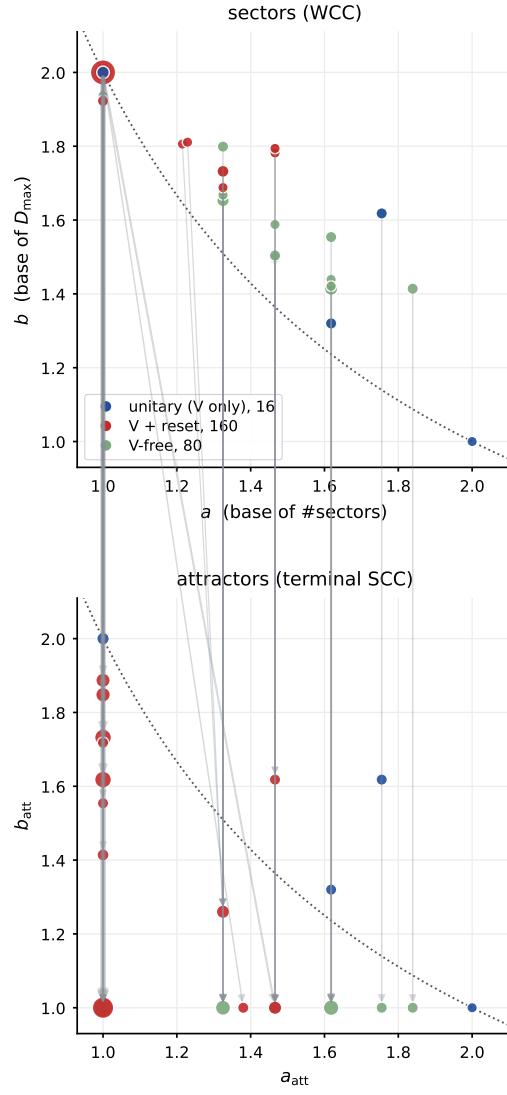


Figure 4: **D** — **twin panels, vertical**. The same, stacked. Fits a single-column format and the downward arrows read naturally as “what dissipation does”, at the cost of a tall aspect ratio.

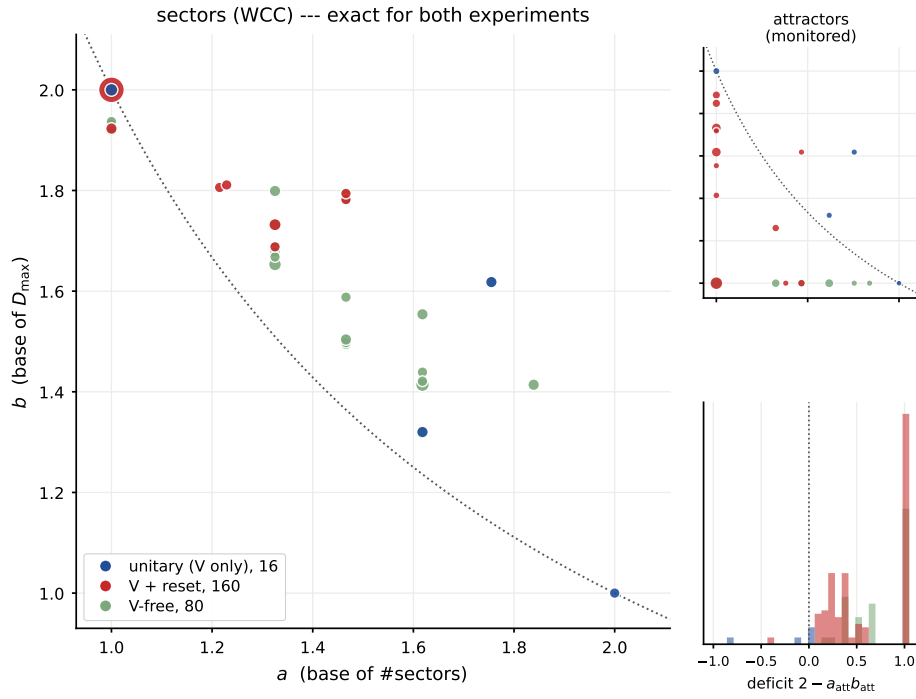


Figure 5: **E** — **inset**. The sector plane full size, the attractor plane as an inset, the deficit distribution below it. Reflects the asymmetry of the two maps rather than pretending they are equals — the sector plane is exact for both experiments, the attractor plane is monitored-only — so it gets less space and a caveat instead of half the figure.

Recommendation

A (overlay) for a journal figure, with **B (slopegraph)** as a panel beside it if space allows.

A is the most compact, and it is the only one that earns its compactness with a finding: putting both coordinates on one set of axes shows that the arrows are overwhelmingly *vertical*. That is not visible in C, D or E, where the two planes are drawn separately and the eye cannot compare positions directly. B then adds the one thing A cannot show cleanly, which is where each rule sits relative to $ab = 2$ before and after.

C and D are the safest if the audience has not seen the two planes before, since nothing has to be explained about hollow-versus-filled markers. E is the right choice only if the asymmetry between the two maps is itself the point of the section.

Reproduce

```
python -m qca_fragmentation.viz.journal --bc obc0
```

writes all five as `figures/fig-journal_{style}_obc0.{pdf,png}`; the module is `code/qca_fragmentation/vi`.